

Computer Security Exam Notes (Slides 1-53)

1. Security

Security protects valuable assets. It is a continuous process, not a final state.

2. Computer vs Internet Security

Computer Security protects local systems. Internet Security protects information transmitted over networks.

3. Cybersecurity

Protects information stored, processed and transmitted in networked systems using policies and technologies.

4. Prevention-Detection-Reaction

Prevention stops attacks, Detection finds attacks, Reaction recovers from attacks.

5. CIA Triad

Confidentiality: keep data secret. Integrity: keep data accurate. Availability: keep systems accessible.

6. Privacy

Users control how personal information is collected and shared.

7. McCumber Cube

3 dimensions: CIA × Information States (Storage, Processing, Transmission) × Measures (Technology, Policy, Education) = 27 cells.

8. Authenticity

Verifies users and data sources are genuine.

9. Accountability

Every action can be traced to the responsible entity using logs and audits.

10. FIPS 199

Impact Levels: Low (limited), Moderate (serious), High (catastrophic).

11. CIA Examples

Confidentiality: student grades. Integrity: hospital allergy records. Availability: authentication server.

12. Computer Security Challenges

Security is complex, attackers need one weakness, requires continuous monitoring, security should be built into design.

13. Assets, Vulnerabilities, Threats, Attacks

Asset = valuable resource. Vulnerability = weakness. Threat = potential danger. Attack = threat exploiting vulnerability. Countermeasures reduce risk.

14. Types of Attacks

Passive: only observes. Active: modifies/disrupts. Insider: internal attacker. Outsider: external attacker.

15. Quick Exam Formulas

Vulnerability → Threat → Attack. CIA = Confidentiality, Integrity, Availability.